

Harrison centre invariance of hypermatrix algebras and representation rings

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Abstract. We establish a hypermatrix algebraic criterion for Harrison centre invariance. Given fusion data, we treat the order-3 fusion tensor as a hypermatrix and use its layer matrices to generate a hypermatrix algebra. If the fusion tensor is fully symmetric and the left-regular multiplication matrices are linearly independent, the Harrison centre of the associated cubic form coincides with the hypermatrix algebra. As an application, we prove that the representation ring of the symmetric group is isomorphic to the corresponding Harrison centre.

Keywords: Harrison centre invariance; hypermatrix algebra; fusion algebra; representation ring; symmetric group.

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1 Introduction

One of the classical questions in invariant theory is to decide when two homogeneous forms of degree at least three are equivalent under an invertible linear change of variables. A standard and effective reduction is to decompose a form as a direct

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sum of subforms with disjoint sets of variables. Such decompositions play a central role across commutative algebra, geometric invariant theory, multilinear algebra, and computational complexity.

In his seminal work [6], Harrison extended Witt’s theory of quadratic forms to higher degrees and showed that, under suitable nondegeneracy hypotheses, every form decomposes uniquely (up to permutation) into indecomposable summands in disjoint sets of variables. The governing algebraic object is the **Harrison centre** of a form f , defined via its Hessian matrix by

$$Z(f) := \{ X \in \text{Mat}_n(\mathbb{K}) \mid H(f)X = X^\top H(f) \}.$$

This finite-dimensional commutative $\mathbb{K}$ -algebra controls direct sum decompositions of f through its idempotents. Since then, centre-type constructions have reappeared in several related contexts [3, 5, 8, 9, 10]. It is convenient to organise the structure constants underlying a form or a monoidal category as entries of an **order- d hypermatrix** (a multiway array). For a cubic form f the coefficients assemble into a symmetric order-3 hypermatrix, and the Hessian $H(f)$ is obtained by a canonical contraction of its full symmetrisation.

The Green (representation) ring of a semisimple tensor category, especially of the representation category of a finite group or a Hopf algebra, records the decomposition rules for tensor products of simples and thus captures both algebraic and monoidal structure. It has been studied in classical settings (group algebras, (quantised) enveloping algebras) and in more exotic examples, including Taft and Sweedler Hopf algebras, Schur algebras, and graded path-coalgebra Hopf algebras [1, 2, 4, 11, 14]. In this paper, we focus on symmetric groups.

In our previous work, we proposed a Harrison centre perspective on the representation ring of symmetric groups and, based on computations, conjectured that the representation ring of $\mathfrak{S}_n$ is isomorphic to the Harrison centre of a naturally attached cubic form [13, Conjecture 18]. Our Harrison–centre criterion identifies $Z(f_{\mathcal{C}})$ with the fusion algebra (equivalently, with the hypermatrix algebra under full symmetry), thereby linking Harrison’s decomposition theory to the fusion data. As an application, we confirm the conjecture for $\text{Rep}(\mathfrak{S}_n)$ by an orthogonal diagonalisation argument based on the normalised character table, without using explicit character values.

In the categorical setting relevant to this paper (see the following sections for more details), the fusion data $\{N_{ij}^k\}$ form an order-3 **fusion hypermatrix**

$\mathcal{N} = (N_{ij}^k)$, whose mode-3 layers $M^{(k)} = (N_{ij}^k)_{i,j}$ generate a natural **hypermatrix algebra**. On the other hand, the left-regular multiplication matrices $L_i = (N_{ij}^k)_{j,k}$ generate the fusion algebra. Under full symmetry, one has $M^{(k)} = (L_k)^\top = L_k$, hence the hypermatrix algebra coincides with the fusion algebra. Our Harrison–centre criterion identifies the commutant of the Hessian with this algebra, thereby linking Harrison’s decomposition theory to a hypermatrix algebra with the fusion data.

The paper is organised as follows. In Section 2, we review basic representation theory, monoidal categories, and Harrison centre theory. In Section 3, we give a sufficient criterion, and apply it to $\text{Rep}(\mathfrak{S}_n)$. We conclude in Section 5 with a conjectural characterisation and several directions for further work.

2 Preliminaries

2.1 Representation theory of finite groups

Throughout this subsection, we work over the field of complex numbers $k = \mathbb{C}$. Let G be a finite group. A (finite-dimensional) complex representation is a homomorphism

$$\rho: G \longrightarrow \text{GL}(V), \quad V \text{ a complex vector space.}$$

Its character is the class function $\chi(g) = \text{Tr}(\rho(g))$.

By Maschke’s theorem, the category $\text{Rep}(G)$ is semisimple. Write a complete set of representatives of isomorphism classes of irreducible representations as $\{\rho_\lambda\}$ with characters $\{\chi_\lambda\}$. The Hermitian inner product on class functions is

$$\langle \chi, \psi \rangle := \frac{1}{|G|} \sum_{g \in G} \chi(g) \overline{\psi(g)}.$$

The irreducible characters form an orthonormal basis with respect to $\langle \cdot, \cdot \rangle$.

For λ, μ , the tensor product decomposes as

$$\rho_\lambda \otimes \rho_\mu \cong \bigoplus_{\nu} c_{\lambda, \mu}^{\nu} \rho_{\nu}, \quad c_{\lambda, \mu}^{\nu} = \langle \chi_\lambda \chi_\mu, \chi_\nu \rangle \in \mathbb{N}.$$

2.2 Harrison–centre theory

Let $\mathbb{K}$ be a field of characteristic 0, and let $f \in \mathbb{K}[x_1, \dots, x_n]$ be a homogeneous polynomial of degree $d \geq 3$. Its **Hessian** is the symmetric matrix

$$H(f) = (\partial_i \partial_j f)_{1 \leq i, j \leq n}, \quad \partial_i := \frac{\partial}{\partial x_i}.$$

A homogeneous form f is called **nondegenerate** if no variable can be removed by an invertible linear change of variables; equivalently, the first partial derivatives $\partial_1 f, \dots, \partial_n f$ are linearly independent.

Following Harrison, the **Harrison centre** of f is the set

$$\begin{aligned} Z(f) &:= \{ X \in \text{Mat}_n(\mathbb{K}) \mid H(f) X = X^\top H(f) \} \\ &= \{ X \in \text{Mat}_n(\mathbb{K}) \mid (H(f) X)^\top = H(f) X \}. \end{aligned}$$

(Here the equality is in $\text{Mat}_n(\mathbb{K}[x_1, \dots, x_n])$.)

Proposition 2.1 ([10, Proposition 2.1]). *Assume that f is nondegenerate. Then the following hold:*

1. *$Z(f)$ is a finite-dimensional commutative $\mathbb{K}$ -subalgebra of $\text{Mat}_n(\mathbb{K})$ containing the identity.*
2. *Direct sum decompositions of f (into summands in disjoint sets of variables, up to an invertible linear change of variables) are in bijection with complete sets of pairwise orthogonal idempotents in $Z(f)$.*
3. *The decomposition of f into a direct sum of indecomposable summands is unique up to equivalence and permutation of summands.*

Remark 2.2. Since $Z(f)$ is finite-dimensional over $\mathbb{K}$, it is an Artin $\mathbb{K}$ -algebra. In particular, it admits a canonical decomposition into local factors $Z(f) = \prod_{i=1}^t Z_i$, and the primitive idempotents of $Z(f)$ are in bijection with these local factors. We will not rely on any topological interpretation of the hypersurface $\{f = 0\}$ here.

2.3 Monoidal categories and fusion hypermatrices

Let $\mathcal{C}$ be a $\mathbb{K}$ -linear semisimple monoidal category with tensor product $\otimes$ and unit object $\mathbf{1}$. We fix associativity and unit constraints and write $X \otimes Y \otimes Z$ without brackets. Let

$$\{X_1, \dots, X_s\} \subset \text{Ob}(\mathcal{C})$$

be a *complete* set of representatives of isomorphism classes of simple objects.

For $1 \leq i, j \leq s$, the tensor product decomposes as

$$X_i \otimes X_j \cong \bigoplus_{k=1}^s N_{ij}^k X_k, \quad N_{ij}^k \in \mathbb{N}.$$

It is convenient to *package* the structure constants into the order-3 **fusion hypermatrix**

$$\mathcal{N} = (N_{ij}^k)_{1 \leq i, j, k \leq s} \in \mathbb{K}^{s \times s \times s}.$$

Two (related but distinct) matrix families extracted from $\mathcal{N}$ will be used:

- the **left regular (multiplication) matrices** $L_i \in \text{Mat}_s(\mathbb{Z})$, defined by

$$(L_i)_{jk} = N_{ij}^k,$$

so that L_i realises left multiplication by $[X_i]$ in the Grothendieck ring;

- the **layer (slice) matrices** $M^{(k)} \in \text{Mat}_s(\mathbb{Z})$, defined by

$$(M^{(k)})_{ij} = N_{ij}^k.$$

We view these integer matrices inside $\text{Mat}_s(\mathbb{K})$ via the natural embedding $\mathbb{Z} \hookrightarrow \mathbb{K}$. In Section 3 we shall work primarily with the left regular matrices $\{L_i\}$.

We say that the fusion **hypermatrix** is **fully symmetric** if

$$N_{ij}^k = N_{ji}^k = N_{ik}^j = N_{ki}^j = N_{jk}^i = N_{kj}^i \quad (\forall i, j, k).$$

This property does not follow from monoidal symmetry alone. In general, monoidal symmetry only implies $N_{ij}^k = N_{ji}^k$; the further equalities above require additional structure. It holds, for example, in $\text{Rep}(\mathfrak{S}_n)$ because all irreducible characters of $\mathfrak{S}_n$ are real-valued (see Section 3).

Definition 2.3. Let $\mathcal{N} = (N_{ij}^k)_{1 \leq i, j, k \leq s}$ be the order-3 fusion hypermatrix. For each k , let $M^{(k)} = (N_{ij}^k)_{1 \leq i, j \leq s}$ be its k -th layer. We define the **hypermatrix algebra** to be the unital $\mathbb{K}$ -subalgebra

$$\mathbf{H}(\mathcal{N}) := \text{Alg}_{\mathbb{K}}(M^{(1)}, \dots, M^{(s)}) \subset \text{Mat}_s(\mathbb{K}).$$

Recall the *fusion algebra*

$$\mathcal{A} := \text{Span}_{\mathbb{K}}\{L_1, \dots, L_s\} \subset \text{Mat}_s(\mathbb{K}), \quad (L_i)_{jk} = N_{ij}^k.$$

Proposition 2.4. *If the fusion hypermatrix is fully symmetric, then $M^{(k)} = (L_k)^\top = L_k$ for all k , and consequently $\mathbf{H}(\mathcal{N}) = \mathcal{A}$.*

Proof. Immediate from $N_{ij}^k = N_{kj}^i$ (equivalently $N_{ij}^k = N_{ik}^j$) and the definitions of $M^{(k)}$ and L_k . $\square$

2.4 Harrison centre invariance

Let $\mathcal{C}$ be a finite $\mathbb{K}$ -linear semisimple monoidal category with simple objects $\{X_1, \dots, X_s\}$ and fusion coefficients N_{ij}^k , and let

$$f_{\mathcal{C}}(x_1, \dots, x_s) = \sum_{i, j, k=1}^s N_{ij}^k x_i x_j x_k$$

be the cubic form naturally attached to this fusion data.

Let $\mathcal{A} := \text{Span}_{\mathbb{K}}\{L_1, \dots, L_s\} \subset \text{Mat}_s(\mathbb{K})$ be the fusion algebra. As will be shown in Theorem 1, the assignment

$$[X_i] \mapsto L_i$$

extends to a surjective $\mathbb{K}$ -algebra homomorphism

$$\Phi: K_0(\mathcal{C}) \otimes_{\mathbb{Z}} \mathbb{K} \longrightarrow \mathcal{A}.$$

Definition 2.5. We say that $\mathcal{C}$ (equivalently, its fusion algebra) has **Harrison centre invariance** if

$$Z(f_{\mathcal{C}}) = \mathcal{A} \subset \text{Mat}_s(\mathbb{K}).$$

Equivalently, $\mathcal{C}$ is Harrison-centre invariant if and only if the map

$$\Phi_{\mathcal{C}}: K_0(\mathcal{C}) \otimes_{\mathbb{Z}} \mathbb{K} \longrightarrow Z(f_{\mathcal{C}}), \quad [X_i] \mapsto L_i,$$

is a $\mathbb{K}$ -algebra isomorphism.

In particular,

$$\begin{aligned} & \Phi_{\mathcal{C}} \text{ is an isomorphism} \\ \iff & \left(Z(f_{\mathcal{C}}) = \text{Span}_{\mathbb{K}}\{L_1, \dots, L_s\} \text{ and } \{L_i\}_{i=1}^s \text{ are linearly independent} \right). \end{aligned}$$

3 Harrison centre invariance

3.1 Invariance in monoidal categories

Let $\mathcal{C}$ be a finite $\mathbb{K}$ -linear semisimple monoidal category with precisely s inequivalent simple objects, labelled $X_1, \dots, X_s$. The tensor product is governed by the non-negative integers N_{ij}^k determined by

$$X_i \otimes X_j \cong \bigoplus_{k=1}^s N_{ij}^k X_k.$$

Package the structure constants into the order-3 fusion hypermatrix $\mathcal{N} = (N_{ij}^k)$. We keep the two associated matrix families:

- the **left regular (multiplication) matrices** $L_i \in \text{Mat}_s(\mathbb{Z})$ with entries $(L_i)_{jk} = N_{ij}^k$, realising left multiplication by $[X_i]$ in the Grothendieck ring;
- the **layer matrices** $M^{(k)} \in \text{Mat}_s(\mathbb{Z})$ with entries $(M^{(k)})_{ij} = N_{ij}^k$, i.e. the mode-3 slices of $\mathcal{N}$.

We view these integer matrices inside $\text{Mat}_s(\mathbb{K})$ via the natural embedding.

Definition 3.1. We say that the fusion *hypermatrix* is **fully symmetric** if

$$N_{ij}^k = N_{ji}^k = N_{ik}^j = N_{ki}^j = N_{jk}^i = N_{kj}^i, \quad \text{for all } 1 \leq i, j, k \leq s.$$

Define the **fusion algebra**

$$\mathcal{A} := \text{Span}_{\mathbb{K}}\{L_1, \dots, L_s\} \subset \text{Mat}_s(\mathbb{K}).$$

Theorem 1. *For all $1 \leq i, j \leq s$, one has*

$$L_i L_j = \sum_{k=1}^s N_{ij}^k L_k.$$

Consequently, the map

$$\Phi: K_0(\mathcal{C}) \otimes_{\mathbb{Z}} \mathbb{K} \longrightarrow \mathcal{A}, \quad [X_i] \longmapsto L_i,$$

is a surjective $\mathbb{K}$ -algebra homomorphism. If $\{L_i\}_{i=1}^s$ are linearly independent, then Φ is an isomorphism.

Proof. By definition, $(L_i)_{jk} = N_{ij}^k$, hence

$$(L_i L_j)_{qk} = \sum_{p=1}^s (L_i)_{qp} (L_j)_{pk} = \sum_{p=1}^s N_{iq}^p N_{jp}^k.$$

Associativity of the tensor product implies

$$\sum_p N_{iq}^p N_{jp}^k = \sum_r N_{ij}^r N_{rq}^k,$$

whence $L_i L_j = \sum_r N_{ij}^r L_r$. The remaining statements are immediate. $\square$

Lemma 3.2. *Assume the fusion hypermatrix is fully symmetric. Then each L_i is a real symmetric matrix, the family $\{L_i\}$ pairwise commutes, and therefore the L_i are simultaneously diagonalisable by a real orthogonal matrix.*

Proof. Full symmetry gives

$$(L_i)_{jk} = N_{ij}^k = N_{ik}^j = (L_i)_{kj},$$

so L_i is symmetric (hence real symmetric, since it has integer entries). Moreover, full symmetry includes $N_{ij}^k = N_{ji}^k$ for all i, j, k , so the Grothendieck ring is commutative and therefore $L_i L_j = L_j L_i$ for all i, j . Commuting real symmetric matrices are simultaneously diagonalisable by an orthogonal matrix. $\square$

For the cubic form

$$f(x_1, \dots, x_s) = \sum_{i,j,k=1}^s N_{ij}^k x_i x_j x_k$$

the Hessian satisfies the following *general* identity:

$$H(x) = (\partial_i \partial_j f)_{1 \leq i, j \leq s} = \sum_{k=1}^s x_k S^{(k)}, \quad S_{ij}^{(k)} := N_{ij}^k + N_{ji}^k + N_{ik}^j + N_{ki}^j + N_{jk}^i + N_{kj}^i. \quad (3.1)$$

Under full symmetry we have $S^{(k)} = 6 M^{(k)}$, and since $M^{(k)} = (L_k)^\top = L_k$, we obtain

$$H(x) = 6 \sum_{i=1}^s x_i L_i. \quad (3.2)$$

Proposition 3.3. *Assume the fusion hypermatrix is fully symmetric and the matrices $\{L_i\}_{i=1}^s$ are linearly independent. Then*

$$\{X \in \text{Mat}_s(\mathbb{K}) \mid L_i X = X^\top L_i \ \forall i\} = \text{Span}_{\mathbb{K}}\{L_1, \dots, L_s\}.$$

Proof. First, since each L_i is symmetric and the L_i commute (Lemma 3.2), any $X \in \text{Span}_{\mathbb{K}}\{L_1, \dots, L_s\}$ is symmetric and commutes with every L_i . Hence $L_i X = X L_i = X^\top L_i$ for all i , so

$$\text{Span}_{\mathbb{K}}\{L_1, \dots, L_s\} \subseteq \{X \mid L_i X = X^\top L_i \ \forall i\}.$$

To prove the reverse inclusion, work first over $\mathbb{R}$. By Lemma 3.2, there exists a real orthogonal matrix U such that $U^\top L_i U = D_i$ is diagonal for all i . Set $X' := U^\top X U$. Then $L_i X = X^\top L_i$ is equivalent to

$$D_i X' = (X')^\top D_i \quad (\forall i),$$

because $U^\top X^\top U = (U^\top X U)^\top = (X')^\top$. Conjugation preserves linear independence, so $\dim_{\mathbb{R}} \text{Span}\{D_i\} = s$. Since the space of diagonal matrices has dimension s , we have

$$\text{Span}_{\mathbb{R}}\{D_1, \dots, D_s\} = \{\text{all diagonal matrices}\}.$$

Therefore $D_i X' = (X')^\top D_i$ for all i is equivalent to

$$D X' = (X')^\top D \quad \text{for all diagonal } D.$$

Writing $X' = (x'_{ab})$, the (a, b) -entry of this identity is $d_a x'_{ab} = d_b x'_{ba}$ for all choices of diagonal scalars $(d_1, \dots, d_s)$. Taking $d_a = 1$ and $d_b = 0$ (with $a \neq b$) forces $x'_{ab} = 0$. Hence X' is diagonal, and the solution space over $\mathbb{R}$ has dimension s .

Finally, the relations $L_i X = X^\top L_i$ form a homogeneous linear system in the entries of X whose coefficients are integers (entries of L_i). Thus the rank of the system (and hence the dimension of its solution space) does not depend on the choice of a characteristic-0 ground field. Consequently, over $\mathbb{K}$ the solution space also has dimension s . Since $\text{Span}_{\mathbb{K}}\{L_1, \dots, L_s\}$ is contained in it and has dimension s , the two spaces coincide. $\square$

Theorem 2. *For the cubic form f above, the Hessian is given by (3.1); under full symmetry it reduces to (3.2).*

Proof. Entrywise,

$$\partial_i \partial_j (x_a x_b x_c) = \delta_{ia} \delta_{jb} x_c + \delta_{ib} \delta_{ja} x_c + \delta_{ia} \delta_{jc} x_b + \delta_{ic} \delta_{ja} x_b + \delta_{ib} \delta_{jc} x_a + \delta_{ic} \delta_{jb} x_a.$$

Summing over (a, b, c) with coefficients N_{ab}^c yields (3.1). If the fusion hypermatrix is fully symmetric, the six summands coincide, whence $S^{(k)} = 6M^{(k)}$ and therefore (3.2). $\square$

Theorem 3. *Assume the fusion hypermatrix is fully symmetric and the matrices $\{L_i\}_{i=1}^s$ are linearly independent. Then $\mathcal{C}$ has Harrison-centre invariance, i.e.*

$$Z(f) = \text{Span}_{\mathbb{K}}\{L_1, \dots, L_s\}.$$

In particular, via Theorem 1, $Z(f) \cong K_0(\mathcal{C}) \otimes_{\mathbb{Z}} \mathbb{K}$ as $\mathbb{K}$ -algebras.

Proof. Under full symmetry, (3.2) gives

$$H(x) = 6 \sum_{i=1}^s x_i L_i.$$

By definition of the Harrison centre,

$$Z(f) = \{ X \in \text{Mat}_s(\mathbb{K}) \mid H(x)X = X^\top H(x) \text{ as polynomials in } x \}.$$

Comparing coefficients of the algebraically independent variables x_i shows that $H(x)X = X^\top H(x)$ holds if and only if

$$L_i X = X^\top L_i \quad (\forall i).$$

Hence

$$Z(f) = \{ X \mid L_i X = X^\top L_i \ \forall i \},$$

and Proposition 3.3 yields

$$Z(f) = \text{Span}_{\mathbb{K}}\{L_1, \dots, L_s\}.$$

The final assertion follows from Theorem 1. $\square$

3.2 Harrison–centre invariant for $\text{Rep}(\mathfrak{S}_n)$

Lemma 3.4 ([12]). *Every finite-dimensional complex representation of the symmetric group $\mathfrak{S}_n$ has a real-valued character, that is, $\chi(g) \in \mathbb{R}$ for all $g \in \mathfrak{S}_n$.*

Theorem 4. *For $\mathfrak{S}_n$, the Kronecker coefficients N_{ij}^k assemble into a totally symmetric third-order hypermatrix in the sense that*

$$N_{ij}^k = N_{ji}^k = N_{ik}^j = N_{ki}^j = N_{jk}^i = N_{kj}^i \quad \text{for all } i, j, k.$$

Proof. By Lemma 3.4 the irreducible characters of $\mathfrak{S}_n$ are real, so $\overline{\chi_\ell(g)} = \chi_\ell(g)$ for every $g \in \mathfrak{S}_n$. Let $\mathcal{K}_1, \dots, \mathcal{K}_s$ be the conjugacy classes, choose representatives $g_t \in \mathcal{K}_t$, set $c_t = |\mathcal{K}_t|$ and write $\chi_{\ell t} := \chi_\ell(g_t)$. The standard character formula gives

$$N_{ij}^k = \frac{1}{|\mathfrak{S}_n|} \sum_{t=1}^s c_t \chi_{it} \chi_{jt} \chi_{kt}.$$

Because every $\chi_{\ell t}$ is real, each summand $c_t \chi_{it} \chi_{jt} \chi_{kt}$ is invariant under any permutation of the indices i, j, k . Hence the stated total symmetry holds. $\square$

As an application, we confirm the Harrison–centre invariance for $\text{Rep}(\mathfrak{S}_n)$ by a direct diagonalisation argument (equivalently, it also follows from Theorems 4 and 3).

Theorem 5. *$\text{Rep}(\mathfrak{S}_n)$ is Harrison–centre invariant.*

Proof. Index the irreducible representations (and characters) by $i = 1, \dots, s$, and the conjugacy classes by $t = 1, \dots, s$ with representatives g_t and sizes c_t . Let $L_i \in \text{Mat}_s(\mathbb{C})$ be the left-regular multiplication matrices,

$$(L_i)_{jk} = N_{ij}^k.$$

By Theorem 4 the fusion hypermatrix is fully symmetric; hence by Theorem 2 the Hessian of

$$f(x_1, \dots, x_s) = \sum_{i,j,k} N_{ij}^k x_i x_j x_k$$

is

$$H(x) = 6 \sum_{i=1}^s x_i L_i.$$

Consider the **normalised character table** $U = (U_{t\ell})_{t,\ell}$ defined by

$$U_{t\ell} = \sqrt{\frac{c_t}{|\mathfrak{S}_n|}} \chi_\ell(g_t) \in \mathbb{R}.$$

The row/column orthogonality relations imply that U is orthogonal: $U^\top U = I = UU^\top$. A standard computation using

$$N_{ij}^k = \frac{1}{|\mathfrak{S}_n|} \sum_{u=1}^s c_u \chi_{iu} \chi_{ju} \chi_{ku}, \quad \sum_{\ell=1}^s \chi_\ell(g_t) \chi_\ell(g_u) = \frac{|\mathfrak{S}_n|}{c_t} \delta_{tu},$$

gives, for each fixed i and all t, j ,

$$(UL_i)_{tj} = \sqrt{\frac{c_t}{|\mathfrak{S}_n|}} \chi_i(g_t) \chi_j(g_t) = (D_i U)_{tj},$$

where $D_i := \text{diag}(\chi_i(g_1), \dots, \chi_i(g_s))$. Hence

$$L_i = U^\top D_i U \quad (1 \leq i \leq s), \quad (3.3)$$

so the L_i are simultaneously diagonalisable over $\mathbb{R}$.

Let $X \in \text{Mat}_s(\mathbb{C})$. By definition of the Harrison centre and the expression $H(x) = 6 \sum_i x_i L_i$, we have

$$\begin{aligned} X \in Z(f) &\iff H(x)X = X^\top H(x) \text{ as polynomials in } x \\ &\iff L_i X = X^\top L_i \quad \forall i. \end{aligned}$$

Conjugating by U and using (3.3), this is equivalent to

$$D_i X' = (X')^\top D_i \quad (\forall i), \quad \text{where } X' := UXU^\top.$$

Since $\{\chi_i\}_{i=1}^s$ form a basis of the class functions, the diagonal algebra is spanned by $\{D_i\}_{i=1}^s$, hence the above is equivalent to

$$DX' = (X')^\top D \quad \text{for all diagonal } D.$$

Writing $X' = (x'_{ab})$, the (a, b) -entry gives $d_a x'_{ab} = d_b x'_{ba}$ for all diagonal scalars. Taking $d_a = 1$ and $d_b = 0$ for $a \neq b$ forces $x'_{ab} = 0$, so X' is diagonal. Therefore X lies in the conjugate of the diagonal algebra, i.e. in

$$U^\top (\text{diagonal algebra}) U.$$

Finally, $\{D_i\}$ are linearly independent, so they form a basis of the diagonal algebra; conjugating back, $\{L_i\}$ are linearly independent and their span equals

$$U^\top(\text{diagonal algebra})U.$$

Hence

$$Z(f) = \text{Span}_{\mathbb{C}}\{L_1, \dots, L_s\}.$$

By Theorem 1, the map $K_0(\text{Rep}(\mathfrak{S}_n)) \otimes_{\mathbb{Z}} \mathbb{C} \rightarrow \text{Span}_{\mathbb{C}}\{L_i\}$, $[X_i] \mapsto L_i$, is an isomorphism. Therefore $\text{Rep}(\mathfrak{S}_n)$ is Harrison-centre invariant. $\square$

4 A sufficient and necessary condition

Throughout this section we specialise to $\mathcal{C} = \text{Rep}(G)$ for a finite group G and work over $\mathbb{K} = \mathbb{C}$. Let $s = |\text{Irr}(G)| = |\text{Cl}(G)|$, and label irreducible characters by $\chi_1, \dots, \chi_s$ and conjugacy classes by representatives $g_1, \dots, g_s$ with sizes $c_t := |[g_t]|$.

Let U be the normalised character table,

$$U_{t\ell} := \sqrt{\frac{c_t}{|G|}} \chi_\ell(g_t) \in \mathbb{C},$$

so U is unitary: $U^*U = I$. Set

$$D_i := \text{diag}(\chi_i(g_1), \dots, \chi_i(g_s)) \quad (1 \leq i \leq s).$$

Then the left-regular multiplication matrices $L_i = (N_{ij}^k)_{j,k}$ satisfy the simultaneous diagonalisation

$$L_i = U^{-1}D_iU \quad (1 \leq i \leq s). \quad (4.4)$$

Since $\{\chi_i\}_{i=1}^s$ form a basis of class functions, the diagonal algebra is spanned by $\{D_i\}_{i=1}^s$. Consequently, the fusion algebra

$$A := \text{Span}_{\mathbb{C}}\{L_1, \dots, L_s\}$$

is conjugate to the full diagonal algebra and hence is a MASA in $\text{Mat}_s(\mathbb{C})$:

$$\text{Cent}_{\text{Mat}_s(\mathbb{C})}(A) = A.$$

Now let $f(x) = \sum_{i,j,k} N_{ij}^k x_i x_j x_k$ be the cubic attached to the fusion data, and write its Hessian as $H(x) = \sum_{k=1}^s x_k S^{(k)}$ as in (3.1).

Definition 4.1. For a subset $\mathcal{T} \subseteq \text{Mat}_s(\mathbb{C})$ we define its **transpose-centraliser** by

$$\text{Cent}^\top(\mathcal{T}) := \{ X \in \text{Mat}_s(\mathbb{C}) \mid TX = X^\top T \text{ for all } T \in \mathcal{T} \}.$$

Lemma 4.2. Let $\mathcal{S} := \{S^{(1)}, \dots, S^{(s)}\}$. Then

$$Z(f) = \bigcap_{k=1}^s \text{Cent}^\top(\{S^{(k)}\}) = \text{Cent}^\top(\text{Span}_{\mathbb{C}} \mathcal{S}).$$

Proof. By definition of the Harrison centre,

$$X \in Z(f) \iff H(x)X = X^\top H(x) \text{ in } \text{Mat}_s(\mathbb{C}[x_1, \dots, x_s]).$$

Since $H(x) = \sum_{k=1}^s x_k S^{(k)}$ and the variables x_k are algebraically independent, this holds if and only if

$$S^{(k)}X = X^\top S^{(k)} \quad (\forall k),$$

which gives the first equality. The second equality follows from linearity: if $S = \sum_k a_k S^{(k)}$, then $SX = \sum_k a_k (S^{(k)}X) = \sum_k a_k (X^\top S^{(k)}) = X^\top S$. $\square$

Theorem 6. Set $A = \text{Span}_{\mathbb{C}}\{L_1, \dots, L_s\}$ and $\mathcal{S} = \{S^{(1)}, \dots, S^{(s)}\}$. Then

$$\text{Rep}(G) \text{ is Harrison-centre invariant} \iff \text{Cent}^\top(\text{Span}_{\mathbb{C}} \mathcal{S}) = A.$$

Proof. By Lemma 4.2, $Z(f) = \text{Cent}^\top(\text{Span}_{\mathbb{C}} \mathcal{S})$. By definition, $\text{Rep}(G)$ is Harrison-centre invariant if and only if $Z(f) = A$. Combining these yields the equivalence. $\square$

Theorem 7. For an abelian finite group G , $\text{Rep}(G)$ is Harrison-centre invariant if and only if $G \cong (\mathbb{Z}/2\mathbb{Z})^r$ (an elementary 2-group).

Proof. Write $\hat{G} = \text{Irr}(G)$ for the character group; then every irreducible representation is 1-dimensional and $\hat{G}$ is an abelian group of order s . Label the basis of $K_0(\text{Rep}(G)) \otimes_{\mathbb{Z}} \mathbb{C}$ by $\{\chi_g\}_{g \in \hat{G}}$. The fusion coefficients are

$$N_{g,h}^k = \delta_{gh,k} \quad (g, h, k \in \hat{G}),$$

so L_g is the permutation matrix of left translation by g on $\hat{G}$.

($\Rightarrow$) Assume $\text{Rep}(G)$ is Harrison–centre invariant, so $Z(f) = A$. In particular, every $L_g \in A$ belongs to $Z(f)$. By Lemma 4.2, this implies $S^{(e)}L_g = L_g^\top S^{(e)}$, where e is the identity of $\hat{G}$. For abelian $\hat{G}$, the explicit formula (3.1) gives

$$S^{(e)} = 4I + 2P,$$

where P is the permutation matrix of inversion $h \mapsto h^{-1}$. Moreover $L_g^\top = L_{g^{-1}}$ and, since $\hat{G}$ is abelian, one has $PL_g = L_{g^{-1}}P$. Hence

$$S^{(e)}L_g = (4I + 2P)L_g = 4L_g + 2PL_g = 4L_g + 2L_{g^{-1}}P,$$

while

$$L_g^\top S^{(e)} = L_{g^{-1}}(4I + 2P) = 4L_{g^{-1}} + 2L_{g^{-1}}P.$$

Equating the two expressions yields $4L_g = 4L_{g^{-1}}$, hence $L_g = L_{g^{-1}}$, and therefore $g = g^{-1}$. Thus every element of $\hat{G}$ has order dividing 2, so $\hat{G} \cong (\mathbb{Z}/2\mathbb{Z})^r$ and hence $G \cong (\mathbb{Z}/2\mathbb{Z})^r$.

($\Leftarrow$) Conversely, if $G \cong (\mathbb{Z}/2\mathbb{Z})^r$, then all irreducible characters are $\{\pm 1\}$ -valued, so the fusion tensor is fully symmetric and the matrices $\{L_i\}$ are linearly independent. Therefore Harrison–centre invariance follows from Theorem 3. $\square$

5 Concluding Remarks

We have established a clear sufficient criterion for Harrison–centre invariance: if the fusion tensor is fully symmetric and the left-regular multiplication matrices $\{L_i\}_{i=1}^s$ are linearly independent, then

$$Z(f_{\mathcal{C}}) = \text{Span}_{\mathbb{K}}\{L_1, \dots, L_s\} \cong K_0(\mathcal{C}) \otimes_{\mathbb{Z}} \mathbb{K}.$$

A conjectural characterisation. The foregoing suggests that the two hypotheses above are not only sufficient but in fact *necessary*. We record this as a precise conjecture.

Conjecture 5.1. Let $\mathcal{C}$ be a finite $\mathbb{K}$ -linear semisimple monoidal category with finitely many simple objects. Then $\mathcal{C}$ has Harrison–centre invariance

$$\left(Z(f_{\mathcal{C}}) \cong K_0(\mathcal{C}) \otimes_{\mathbb{Z}} \mathbb{K} \text{ via } [X_i] \mapsto L_i \right)$$

if and only if

- (i) its fusion tensor (N_{ij}^k) is fully symmetric, and
- (ii) the left-regular matrices L_i (defined by $(L_i)_{jk} = N_{ij}^k$) are linearly independent, equivalently $\dim_{\mathbb{K}} \text{Span}\{L_1, \dots, L_s\} = s$.

If Conjecture 5.1 holds, one obtains a clean dichotomy: finite semisimple monoidal categories split into a Harrison–centre invariant class—those with fully symmetric, independent left-regular layers—and a complementary class failing at least one of these algebraic tests.

Future directions. We highlight a few directions for further investigation:

1. Verifying the conjecture for broader families (quantum groups at roots of unity, Deligne categories, wreath products $C_m \wr \mathfrak{S}_n$, and Weyl groups of other types), and identifying minimal hypotheses replacing full symmetry when appropriate.
2. Analysing the failure modes of invariance: how lack of full symmetry or of linear independence manifests in the joint diagonalisation picture and in representation-theoretic or geometric features.
3. Algorithmic aspects: given fusion data (N_{ij}^k) , deciding Harrison–centre invariance via numerical joint diagonalisation of $\{L_i\}$ and dimension tests for $\text{Span}\{L_i\}$.

We hope that the present sufficient criterion, together with the conjectural characterisation above, offers a useful organising principle for the Harrison–centre behaviour of monoidal categories.

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